Tutorial 7

26 December 2023

Show that if $a \mid b$ and $b \mid a$, where a and b are integers, then $a = b$ or $a = -b$.

$a \mid b \Rightarrow b = an$ for some $n \in \mathbb{Z}$

$b \mid a \Rightarrow a = kb$ for some $k \in \mathbb{Z}$

$\Rightarrow ab = abkn \Rightarrow ab(1 - kn) = 0$

$\Rightarrow kn = 1$ since $a \neq 0, b \neq 0$

Thus, $k = 1, n = 1$
or $k = -1, n = -1$

Show that if a, b, c, and d are integers, where $a \neq 0$, such that $a \mid c$ and $b \mid d$, then $ab \mid cd$.

Since $a \mid c \Rightarrow c = an$ for some $n \in \mathbb{Z}$

$b \mid d \Rightarrow d = kb$ " $k \in \mathbb{Z}$

$\Rightarrow cd = ab(kn) \Rightarrow ab \mid cd$

Show that if a, b, and c are integers, where $a \neq 0$ and $c \neq 0$, such that $ac \mid bc$, then $a \mid b$.

$ac \mid bc \Rightarrow bc = kac \Rightarrow b = ka$

$\Rightarrow a \mid b.$

Prove or disprove that if $a \mid bc$, where a, b, and c are positive integers and $a \neq 0$, then $a \mid b$ or $a \mid c$.

let $a = 4$ $b = 14$, $a = 10$

False

Prove that if a and b are integers and a divides b, then a is odd or b is even.

$a|b \Rightarrow$ a is odd or b is even.

by counterposition

suppose a is even and b is odd

$\Rightarrow$ $a = 2k$ for some k

$b = 2n + 1$ for some n

$b/a = \frac{2n}{2k} + \frac{1}{2k} \notin \mathbb{Z} \Rightarrow a \nmid b.$

Prove that if a and b are nonzero integers, a divides b, and $a + b$ is odd, then a is odd.

$a|b \Rightarrow b = na$, $n \in \mathbb{Z}$

$a + b = 2k + 1$, $k \in \mathbb{Z}$

proof by contradiction.

if a is even.

$a = 2b$

$\Rightarrow \quad a+b = (1+n)a = 2L(1+n)$ which is a contradiction.

Prove that if a is an integer that is not divisible by 3, then $(a + 1)(a + 2)$ is divisible by 3.

Since $a \nmid 3 \Rightarrow a = 3q + r, \quad r \neq 0$
$0 < r < 3$
r can be ①, ②

$\Rightarrow$ case ① $a+1 = 3q + 2$
$a+2 = 3q+3 = 3(q+1)$
$\Rightarrow 3 \mid (a+1)(a+2)$
Similarly for case ②.

Prove that if a is a positive integer, then 4 does not divide $a^2 + 2$.

since $\mathbb{Z}_4 = \{0, 1, 2, 3\}$

by cases

- $a \equiv 0 \Rightarrow a^2 = 0 \qquad 2 \equiv 2$
  $\Rightarrow a^2+2 \equiv 2.$
- $a \equiv 1 \Rightarrow a^2 = 1 \Rightarrow a^2+2 \equiv 3$
- $a \equiv 2 \Rightarrow a^2 \equiv 4 \equiv 0 \Rightarrow a^2+2 \equiv 2.$
- $a \equiv 3 \Rightarrow a^2 \equiv 9 \equiv 1 \Rightarrow a^2+2 \equiv 3.$

What time does a 24-hour clock read

What time does a 24-hour clock read

a) 100 hours after it reads 2:00?

b) 45 hours before it reads 12:00?

c) 168 hours after it reads 19:00?

a) $100 = 4(24) + 4$

$\Rightarrow 100 \equiv 4 \pmod{24}$

$2 \equiv 2 \pmod{24}$

$\Rightarrow 102 \equiv 6 \pmod{24}$

b) $12 \equiv 12 \pmod{24}$

$45 \equiv 21 \pmod{24}$

$\Rightarrow -33 \equiv -9 \equiv 15 \pmod{24}$

$-9 = 24 + 15$

c) $168 \equiv 0 \pmod{24}$

Suppose that a and b are integers, $a \equiv 11 \pmod{19}$, and $b \equiv 3 \pmod{19}$. Find the integer c with $0 \leq c \leq 18$ such that

a) $c \equiv 13a \pmod{19}$.

b) $c \equiv 8b \pmod{19}$.

c) $c \equiv a - b \pmod{19}$.

d) $c \equiv 7a + 3b \pmod{19}$.

e) $c \equiv 2a^2 + 3b^2 \pmod{19}$.

f) $c \equiv a^3 + 4b^3 \pmod{19}$.

a) $a \equiv 11$, $b \equiv 3$

$c \equiv 13a$

since $13 \equiv 13$, $a \equiv 11$ $\Rightarrow$ $13a \equiv 143 \equiv \boxed{10}$

b) $c \equiv 8b$ $\quad 8 \equiv 8$, $b \equiv 3$

$\Rightarrow 8b \equiv 24 \equiv \boxed{5}$

c) $c \equiv a - b \equiv 11 - 3 \equiv 8$

f) $a^3 \equiv 1331 \equiv 1$ $\quad 4b^3 \equiv 4 \times 8 \equiv 13$

$b^3 \equiv 27 \equiv 8$

$a^3 + 4b^3 = 1 + 13 = 14$

Show that if a and d are positive integers, then $(-a)$ **div** $d = -a$ **div** d if and only if d divides a.

$\Rightarrow$ $(-a) \text{ div } d = -a \text{ div } d$

Let $a \text{ div } d = q$

$\Rightarrow a = qd + r_1$

and

$-a = -qd + r_2$

$\Rightarrow r_1 + r_2 = 0 \quad \Rightarrow r_1 = -r_2$

since $r \geq 0$

$\Rightarrow r_1 = r_2 = 0 \Rightarrow d \mid a$

$\Leftarrow$ $d \mid a \Rightarrow a = dn$

$(-a) = (-n)d$

$\Rightarrow (-a) \text{ div } d = -a \text{ div } d$ $\quad \#$

Prove or disprove that if a, b, and d are integers with $d > 0$, then $(a + b)$ **div** $d = a$ **div** $d + b$ **div** d.

False Let $a = k_1 d + r_1$

$b = k_2 d + r_2$

$\Rightarrow \quad a+b = (k_1+k_2)\,d + (r_1+r_2)$

if $r_1 + r_2 > d$

Then the statement is false

Counter example,

$d = 2$

$a = 3 \quad , \quad b = 9$

$3 = (1)\,2 + 1$

$9 = (4)\,2 + 1$

$3+9 = (5)\,2 + 2 = (6)\,2 + 0$

$\Rightarrow \quad (3+9)\ \text{div}\ 2 = 6 \neq 5 = 3\ \text{div}\ 2 + 9\ \text{div}\ 2$

Let m be a positive integer. Show that $a \equiv b \pmod{m}$ if a **mod** $m = b$ **mod** m.

$a \equiv b \pmod{m} \Rightarrow m \mid (a-b) \Rightarrow a - b = mn.$

Let $a = mk_1 + r_1$

$b = mk_2 + r_2$

$\Rightarrow \quad a - b = (k_1 - k_2)\,m + (r_1 - r_2)$

$= mn$

Since $r_1, r_2 < m$ by definition

$\Rightarrow r_1 - r_2 < m$

$\Rightarrow \quad r_1 = r_2 \quad \Rightarrow \quad a \bmod m = b \bmod m.$

Another proof:

$a \equiv a \bmod m \pmod m$

$b \equiv b \bmod m \pmod m$

since $a, b \in$ to same congruence class i.e

$a \equiv b \Rightarrow a \bmod m \equiv b \bmod m$

$\Rightarrow a \bmod m - b \bmod m = mn$

since $a \bmod m < m$

$b \bmod m < m \Rightarrow a \bmod m - b \bmod m < m$

$\Rightarrow n = 0$ proving the statement.

Find the integer a such that

a) $a \equiv 43 \pmod{23}$ and $-22 \leq a \leq 0$.

b) $a \equiv 17 \pmod{29}$ and $-14 \leq a \leq 14$.

c) $a \equiv -11 \pmod{21}$ and $90 \leq a \leq 110$.

a) $23 \mid (43 - a)$

$\Rightarrow 43 - a = 23n \qquad n = 2$

$a = 43 - 46 = -3$

b) $a = 17 - n(29) = -12$

c) $a = -11 - n(21) = 94$

$n = (-5)$

Find each of these values.

a) $(177 \bmod 31 + 270 \bmod 31) \bmod 31$

b) $(177 \bmod 31 \cdot 270 \bmod 31) \bmod 31$

b) $(177 \bmod 31 + 270 \bmod 31) \bmod 31$

$= (177 \times 270) \bmod 31 = 47790 \bmod 31$

$= 19$

a) $447 \bmod 31 = 13$

Show that if $a \equiv b \pmod{m}$ and $c \equiv d \pmod{m}$, where a, b, c, d, and m are integers with $m \geq 2$, then $a - c \equiv b - d \pmod{m}$.

$\Rightarrow m \mid a-b$ & $m \mid c-d$

$\Rightarrow a - b = mn_1$, $c - d = mn_2$ $\Rightarrow (a-c) - (b-d) = (n_1 - n_2)m$

$\Rightarrow m \mid (a-c) - (b-d) \Rightarrow a - c \equiv b - d$

Show that if $n \mid m$, where n and m are integers greater than 1, and if $a \equiv b \pmod{m}$, where a and b are integers, then $a \equiv b \pmod{n}$.

$n \mid m \Rightarrow m = k_1 n$

since $a \equiv b \pmod{m} \Rightarrow m \mid a-b$

$\Rightarrow \quad a-b = k_2 m = k_1 k_2 n$

$\Rightarrow \quad n \mid a-b \Rightarrow \quad a \equiv b \pmod{n}$.

Show that if a, b, c, and m are integers such that $m \geq 2$, $c > 0$, and $a \equiv b \pmod{m}$, then $ac \equiv bc \pmod{mc}$.

$a \equiv_m b \Rightarrow m \mid b-a \Rightarrow b-a = mn$

multiply by c

$\Rightarrow \quad c(b-a) = mnc$

$\Rightarrow \quad mc \mid bc - ac \Rightarrow \quad ac \equiv bc \pmod{mc}$

Use Exercise 44 to show that if m is a positive integer of the form $4k + 3$ for some nonnegative integer k, then m is not the sum of the squares of two integers.

if $n \in \mathbb{Z} \Rightarrow n^2 \bmod 4 = (0, 1)$

$m = 4K+3 \quad K \geq 0 \Rightarrow m \neq a^2 + b^2$

by counterposition

$m = a^2 + b^2$ since $a^2, b^2 \bmod 4$ can only be $1, 0$

$m \bmod 4 = (a^2+b^2) \bmod 4 = (0, 1, 2)$

Thus, $m = 4K+0$ or $4K+1$ or $4K+2$ it cannot be (3).

Prove that if n is an odd positive integer, then $n^2 \equiv 1 \pmod 8$.

$n = 2m+1 \Rightarrow n^2 = 4m^2 + 4m + 1$
$= 4m(m+1) + 1$

since either (m) or $(m+1)$ is even
$\Rightarrow 8 \mid 4m(m+1) \Rightarrow n^2 \bmod 8 = 1 \quad m \geq 1$

$m = 0$ True by definition.

Show that if a, b, k, and m are integers such that $k \geq 1$, $m \geq 2$, and $a \equiv b \pmod m$, then $a^k \equiv b^k \pmod m$.

$a \equiv b \qquad a \equiv a \quad , b \equiv b$

$\Rightarrow a^2 \equiv ab \equiv b^2 \Rightarrow a^3 \equiv a^2 b \equiv b^2 a$
$\equiv b^3$

by induction, $a^k \equiv b^k$.

Determine whether each of these integers is prime.

a) 19 **b)** 27
c) 93 **d)** 101
e) 107 **f)** 113

a) $\lfloor \sqrt{19} \rfloor = 4$ 19/2, 19/3

c) $\lfloor \sqrt{93} \rfloor = 9$, $\{2, 3, 5, 7\}$

$93 = 31 \times 3$.

Find the prime factorization of each of these integers.

a) 39 **b)** 81 **c)** 101
d) 143 **e)** 289 **f)** 899

a) $\lfloor 39 \rfloor = 6$ $\{2, 3, 5\}$

$39 = 3 \times 13$

f) $\lfloor 899 \rfloor = 29$ $\{2, 3, 5, 7, 11, 13, 17, 19, 23, 29\}$

$899 = 31 \times 29$

Find the prime factorization of 10!.

$10! = 10 \times 9 \times 8 \times 7 \times 6 \times 5 \times 4 \times 3 \times 2$

$= [7 \times 5 \times 3 \times 2][2 \times 5][3 \times 3][2^3]$

$[2 \times 3][2^2]$

$= 7 \times 5^2 \times 3^4 \times 2^8$

How many zeros are there at the end of 100!?

Zero are produced by multiplication of $10 = (2, 5)$
$100!$ contains

So, we count factors of 2, 5 since
factors 2 > factors of 5

we need only to count factors of 5.

5, 10, 15, 20, 25, ..., 100

$10 \times 2 = 20$ factor but 25, 50, 75, 100 has one more factor of 5, $\Rightarrow$ 24 zero.

Show that $a^m + 1$ is composite if a and m are integers greater than 1 and m is odd. [*Hint:* Show that $x + 1$ is a factor of the polynomial $x^m + 1$ if m is odd.]

a, m m is odd $\Rightarrow$ a^m+1 is composite.

$m = 2n+1$

$\Rightarrow \quad a^{2n+1}+1$

$$a^{2n+1}+1 = (a+1)(a^{2n} - a^{2n-1} + a^{2n-2} - \cdots + 1)$$

$a^{2n+1} - a^{2n} + a^{2n-1} - \cdots + a$
$a^{2n} - a^{2n-1} + \cdots + 1$

Thus, (a+1) is a fact for proving the result.

Show that if $2^m + 1$ is an odd prime, then $m = 2^n$ for some nonnegative integer n. [*Hint:* First show that the polynomial identity $x^m + 1 = (x^k + 1)(x^{k(t-1)} - x^{k(t-2)} + \cdots - x^k + 1)$ holds, where $m = kt$ and t is odd.]

$$x^m + 1 = (x^k + 1)(x^{k(t-1)} - x^{k(t-2)} + \cdots + 1)$$

$m = kt$ multiply RHS to see equivalence.

now for the proof: we want to show that $m = 2^n$
Thus, making m even with no primes other than 2.

proof by contradiction, suppose that has another prime fator $t > 2$ such that $m = kt$.
Thus,

$$2^m + 1 = (2^k + 1)(\cdots)$$

however, $2^m + 1$ is an odd prime Thus, $(\cdots)$ must be
① $\Rightarrow$ $m = k \Rightarrow t = 1$ which is a contradiction.

Show that $\log_2 3$ is an irrational number. Recall that an irrational number is a real number x that cannot be written as the ratio of two integers.

proof by contradiction
suppose

$$\log_2 3 = \frac{a}{b}$$

$$\Rightarrow \quad b \log_2 3 = a$$

$$\cdots \quad 3^b = 2^a$$

$\Rightarrow$ $\log_2 3^b = a$

$\Rightarrow$ $2^a = 3^b$ which cannot be true by fundamental thm of arithmetic

Prove that for every positive integer n, there are n consecutive composite integers. [*Hint:* Consider the n consecutive integers starting with $(n + 1)! + 2$.]

Using the hint consider

$(n+1)! + 2$ This is divisible by 2

↑ includes ②

Similarly $3 \mid (n+1)! + 3$

$4 \mid (n+1)! + 4$

$\vdots$

$n \mid (n+1)! + n$

$(n+1) \mid (n+1)! + (n+1)$

Thus, $\exists$ n consecutive composite integers.

Prove or disprove that there are three consecutive odd positive integers that are primes, that is, odd primes of the form p, $p + 2$, and $p + 4$.

2, 3, 5, 7 $p=3$, $p+2=5$, $p+4=7$

Determine whether the integers in each of these sets are pairwise relatively prime.

a) 21, 34, 55 **b)** 14, 17, 85

c) 25, 41, 49, 64 **d)** 17, 18, 19, 23

a) $\gcd(21,34) = \gcd(21,13) = \gcd(13,8) = \gcd(8,5)$
$= \gcd(5,3) = \gcd(3,2) = 1$

$\gcd(55,34) = \gcd(34,21) = 1$
$\gcd(55,21) = \gcd(21,13) = 1$

What are the greatest common divisors of these pairs of integers?

a) $3^7 \cdot 5^3 \cdot 7^3, 2^{11} \cdot 3^5 \cdot 5^9$
b) $11 \cdot 13 \cdot 17, 2^9 \cdot 3^7 \cdot 5^5 \cdot 7^3$
c) $23^{31}, 23^{17}$
d) $41 \cdot 43 \cdot 53, 41 \cdot 43 \cdot 53$
e) $3^{13} \cdot 5^{17}, 2^{12} \cdot 7^{21}$
f) $1111, 0$

a) $\gcd = p_1^{\min(a_1,b_1)} \cdots p_n^{\min(a_n,b_n)}$

$= 2^0 3^5 5^3 7^0$

f) $\gcd(1111, 0) = 1111$

Find gcd(1000, 625) and lcm(1000, 625) and verify that $\gcd(1000, 625) \cdot \text{lcm}(1000, 625) = 1000 \cdot 625$.

$\gcd(1000, 625) = \gcd(625, 375) = \gcd(375, 250)$
$= \gcd(250, 125) = \gcd(125, 0) = 125$

$\text{lcm}(1000, 625) = 5000$

Show that if a and b are positive integers, then $ab = \gcd(a, b) \cdot \text{lcm}(a, b)$. [*Hint:* Use the prime factorizations of a and b and the formulae for $\gcd(a, b)$ and $\text{lcm}(a, b)$ in terms of these factorizations.]

let $a = p_1^{a_1} \cdots p_n^{a_n}$

$b = p_1^{b_1} \cdots p_n^{b_n}$

Note that either $a_i > b_i$, $a_i = b_i$ or $a_i < b_i$

since $\gcd = \prod_i p_i^{\min(a_i, b_i)}$

$\text{lcm} = \prod_i p_i^{\max(a_i, b_i)}$

$\Rightarrow \prod_i p_i^{\min(a_i, b_i)} p_i^{\max(a_i, b_i)}$

$= \prod_i p_i^{a_i} p_i^{b_i} = ab.$

Using the method followed in Example 17, express the greatest common divisor of each of these pairs of integers as a linear combination of these integers.

a) 9, 11 **b)** 33, 44 **c)** 35, 78
d) 21, 55 **e)** 101, 203 **f)** 124, 323
g) 2002, 2339 **h)** 3457, 4669 **i)** 10001, 13422

e) $\gcd(101, 203) = (s)\,101 + (t)\,203$

$203 = (2)\,101 + 1$

$1 = 203 - 2(101)$

f)

$323 = 2(124) + 75$ $\quad 75 = 323 - 2(124)$

$124 = 75 + 49$ $\quad 49 = 124 - 75$

$75 = 49 + 26$ $\quad 26 = 75 - 49$

$49 = 26 + 23$ $\quad 23 = 49 - 26$

$26 = 23 + 3$ $\quad 3 = 26 - 23$

$23 = 7(3) + 2$ $\quad 2 = 23 - 7(3)$

$26 = 23 + 3$ $\quad 3 = 26 - 25$

$23 = 7(3) + 2$ $\quad 2 = 23 - 7(3)$

$3 = 2 + 1$ $\quad 1 = 3 - 2$

$1 = 3 - 2$

$= 3 - 23 + 7 \times 3 = 8 \times 3 - 23$

$= 8 \times 26 - 8 \times 23 - 23 = 8 \times 26 - 9 \times 23$

$= 8 \times 26 - 9 \times 49 + 9 \times 26 = 17 \times 26 - 9 \times 49$

$= 17 \times 75 - 17 \times 49 - 9 \times 49$

$= 17 \times 75 - 26 \times 49$

$= 17 \times 75 - 26 \times 124 + 75 \times 26$

$= 43 \times 75 - 26 \times 124$

$= 43 \times 323 - 86(124) - 26 \times 124$

$= 43 \times 323 - 112(124).$

Use the extended Euclidean algorithm to express gcd(252, 356) as a linear combination of 252 and 356.

$356 = (1)(252) + 104$ $\quad$ 1 0

$252 = (2)(104) + 44$ $\quad$ 0 1

$104 = (2)(44) + 16$ $\quad$ 1 −1

$44 = (2)(16) + 12$ $\quad$ −2 3

$16 = (1)12 + 4$ $\quad$ 5 −7

$12 = (3)4 + 0$ $\quad$ −12 17

17 −24

$17 \times 356 - 24 \times 252 = 4.$

Find the smallest positive integer with exactly n different positive factors when n is

a) 3. **b)** 4. **c)** 5.
d) 6. **e)** 10.

Consider

$n = p_1 p_2$

factors $1, p_1, p_2, p_1$

$n = p_1^2 p_2$

$(1, p_1, p_1^2)(p_2, p_1 p_2, p_1^2 p_2)$

$n = p_1^2 p_2^2$

$\left\{ \begin{matrix} (1, p_1, p_1^2) \\ (1, p_1 p_2, p_1^2 p_2) \\ (p_2^2, p_1 p_2^1, p_1^2 p_2^2) \end{matrix} \right\}$ $(2+1)$ $\Big| (2+1)$

$n = p_1^{e_1} p_2^{e_2}$

$(e_1 + 1)(e_2 + 2)$

$\Rightarrow \quad n = p_1^{e_1} p_2^{e_2} \cdots p_n^{e_n}$

it has $(e_1+1)(e_2+1) \cdots (e_n+1)$ factors

a) $(e_1+1)(e_2+1) \cdots (e_n+1) = 3$ prime

$\Rightarrow e_i \quad i > 1 \quad = 0$

$\Rightarrow e_1 = 2$

$2^2 = 4.$

b) $(e_1+1)(e_2+1) \cdots (e_n+1) = 2 \times 2$

$e_1 = e_2 = 1$

$n = 2 \times 3 = 6$

Prove that the product of any three consecutive integers is divisible by 6.

$(n+1)(n+2)(n+3)$ we want to check if 6 divides this number

by prime factorization $6 = 2 \times 3$

For 6 to divide this number it must contain a factor of 2 and a factor of 3.

Since we have 3 consecutive numbers a factor of two must exist

{ odd even odd
even odd even. }

So, we want to check if a set of 3 exists!

Since $\mathbb{Z}_3 = \{0, 1, 2\}$ one of the three numbers must be divisible by 3 proving the statement.

Show that if a, b, and m are integers such that $m \geq 2$ and $a \equiv b \pmod{m}$, then $\gcd(a, m) = \gcd(b, m)$.

$\Rightarrow \quad m \mid a - b \quad$ or $\quad a = b + mn.$

$a \bmod m = b \bmod m = r$

$a = k_1 m + r$

$b = k_2 m + r$

let $d = \gcd(b, m) \Rightarrow d \mid a$

similarly for any divisor of a, m thus,

$\gcd(a, m) = \gcd(b, m)$

Adapt the proof in the text that there are infinitely many primes to prove that there are infinitely many primes of the form $3k + 2$, where k is a nonnegative integer. [*Hint:* Suppose that there are only finitely many such primes $q_1, q_2, \ldots, q_n$, and consider the number $3q_1q_2 \cdots q_n - 1$.]

suppose ∃ finitely many primes of this form

Consider

$Q = 3q_1 \cdots q_n - 1$

First notice that all primes of the form $3k+2$ cannot divide Q nor 3 since ∃ a remainder of (-1).

- If Q is prime then we are done
- If Q is not a prime

$Q = p_1^- p_2^- \cdots p_n^-$

non of the p_s can be 3 since $Q \bmod 3 = -1$

thus, p_s can only have the form

$(3k+1)$ with 1 as a remainder.

since $(3k_1+1)(3k_2+1) = 3(3k_1k_2 + k_1 + k_2) + 1$

$\Rightarrow$ Q must also have the form $(3k+1)$

$Q \bmod 3 = 1$

however, $Q \bmod 3 = 2$ Thus, ∃ a prime of the form $(3k+2)$ not in the list which is multiplied as a factor of Q proving the statment.